

CHAPTER 1

THE LIVING WORLD



1.1 Diversity in the Living World

1.2 Taxonomic Categories

Kitna amazing hai ye living world! Living beings ki variety really mind-blowing hai. Humein alag-alag habitats mein living organisms milte hain – chahe wo thande pahad ho, ghane jungle ho, samundar ho, fresh water lakes ho, deserts ho ya garam paani ke springs. Ek daudta hua ghoda, migrate karte birds ka lamba safar, phoolon se bhari valley ya shikaar karta shark – ye sab humein hairan kar dete hain aur nature ke liye ek deep sense of wonder feel hota hai. Ek population ke members ke beech ka ecological conflict aur cooperation, different populations ke beech ka relation, ya ek cell ke andar molecules ka lagataar movement – ye sab humein sochne par majboor karte hain: life aakhir hai kya? Is chapter mein do important questions chhipe hue hain. Pehla technical question hai – living aur non-living mein difference kya hai? Doosra philosophical question hai – life ka purpose kya hai? Scientists ke roop mein hum doosre question ka answer nahi denge. Is chapter mein hum pehle question par focus karenge – what is living?

1.1 DIVERSITY IN THE LIVING WORLD

Agar tum apne aas-paas dhyan se dekho, to tumhe bahut saare living organisms dikhenge – jaise potted plants, insects, birds, tumhare pets, aur aur bhi kai animals aur plants. Aise bhi kai organisms hote hain jo tumhe naked eye se nahi dikhte, lekin wo bhi tumhare around hi hote hain. Agar tum apni observations ka area badha do, to tumhe aur bhi zyada variety aur range ke organisms dekhne ko milenge. Obviously, agar tum kisi dense forest mein jaoge, to wahan tumhe living organisms ki sankhya aur unke kinds bahut zyada dikhenge. Har alag plant, animal ya organism jo tum dekhte ho, ek **species** ko represent karta hai. Ab tak jo species describe ki gayi hain, unki sankhya lagbhag 1.7 – 1.8 million ke beech hai. Isse hum **biodiversity** kehte hain, yani Earth par maujood organisms ki number aur variety.

types of organisms present on earth. We should remember here that as we explore new areas, and even old ones, new organisms are continuously being identified.

As stated earlier, there are millions of plants and animals in the world; we know the plants and animals in our own area by their local names. These local names would vary from place to place, even within a country. Probably you would recognise the confusion that would be created if we did not find ways and means to talk to each other, to refer to organisms we are talking about.

Hence, there is a need to standardise the naming of living organisms such that a particular organism is known by the same name all over the world. This process is called **nomenclature**. Obviously, nomenclature or naming is only possible when the organism is described correctly and we know to what organism the name is attached to. This is **identification**.

In order to facilitate the study, number of scientists have established procedures to assign a scientific name to each known organism. This is acceptable to biologists all over the world. For plants, scientific names are based on agreed principles and criteria, which are provided in International Code for Botanical Nomenclature (ICBN). You may ask, how are animals named? Animal taxonomists have evolved International Code of Zoological Nomenclature (ICZN). The scientific names ensure that each organism has only one name. Description of any organism should enable the people (in any part of the world) to arrive at the same name. They also ensure that such a name has not been used for any other known organism.

Biologists follow universally accepted principles to provide scientific names to known organisms. Each name has two components – the **Generic name** and the **specific epithet**. This system of providing a name with two components is called **Binomial nomenclature**. This naming system given by Carolus Linnaeus is being practised by biologists all over the world. This naming system using a two word format was found convenient. Let us take the example of mango to understand the way of providing scientific names better. The scientific name of mango is written as *Mangifera indica*. Let us see how it is a binomial name. In this name *Mangifera* represents the genus while *indica*, is a particular species, or a specific epithet. Other universal rules of nomenclature are as follows:

1. Biological names are generally in Latin and written in italics. They are Latinised or derived from Latin irrespective of their origin.
2. The first word in a biological name represents the genus while the second component denotes the specific epithet.
3. Both the words in a biological name, when handwritten, are separately underlined, or printed in italics to indicate their Latin origin.